

EE371 Term Project

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1. Modulation and Demodulation of a Single-Tone Message

All signal creation and processing were carried out in MATLAB. The first step was to determine a suitable sampling frequency f_s to create a time vector. A larger f_s provides more accurate results but at the cost of increased computational requirements and memory usage. Considering the Nyquist criterion, I chose $f_s = 100$ MHz, which is sufficiently larger than $2f_c$. The time vector was defined as $t = [0 : \frac{1}{f_s} : d]$. Additionally, the duration of the time signal was set as $d = 1 \text{ s} \gg \frac{1}{f_s}$.

1.1. Creating the Sine Wave

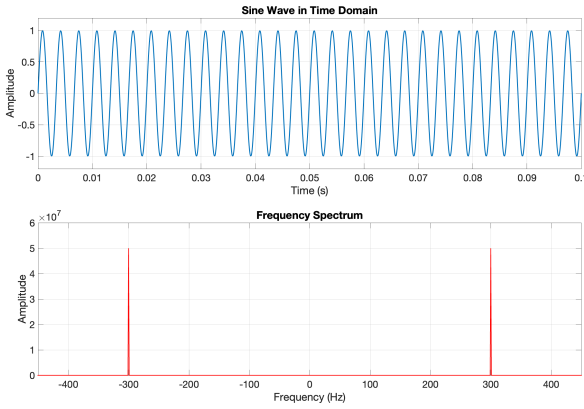


Figure 1: 300 Hz sine wave.

A sinusoidal wave with an amplitude of 1 and a period of 0.0033 s was generated. Its frequency spectrum exhibits two Dirac pulses located at +300 Hz and -300 Hz, as expected.

1.2. AM Modulation

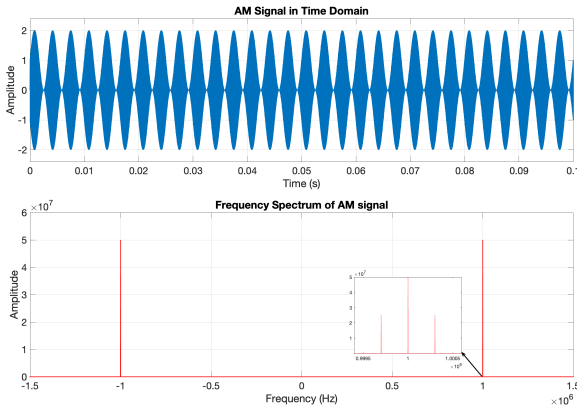


Figure 2: AM modulated signal.

For amplitude modulation with $\mu = 1$, a DC signal of magnitude 1 was added to the original message signal, corresponding to the time vector (all signals are arrays of length f_s).

This signal was then multiplied by a carrier wave of frequency $f_c = 10^6$, defined as $\sin(2\pi f_c t)$.

In the time domain, the modulated signal clearly shows a 300 Hz envelope. In the frequency domain, the baseband signal, consisting of a Dirac pulse at 0 Hz and two symmetric pulses of the sine wave, is shifted to the carrier frequency of 10^6 Hz.

1.3. Adding Noise to the AM Modulated Signal

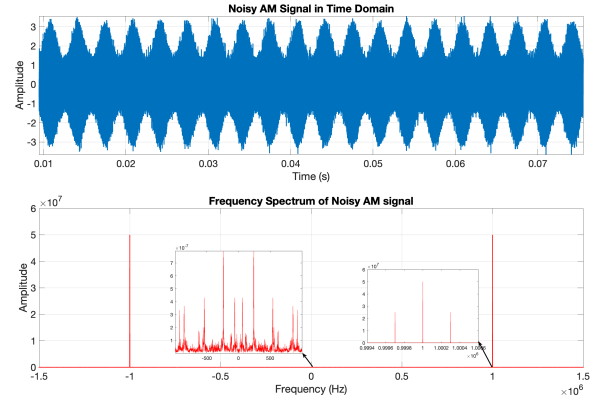


Figure 3: Extracted voice signal.

To simulate an AWGN channel, a normally distributed random variable was generated using the 'randn()' function and scaled by $\sigma = 0.04$. Adding this noise to the AM-modulated signal caused significant distortion in the time-domain waveform. However, the spectral noise density around the baseband was much lower (on the order of 10^{-7}), indicating that recovering the original signal should still be feasible.

1.4. Recovering the Original Signal

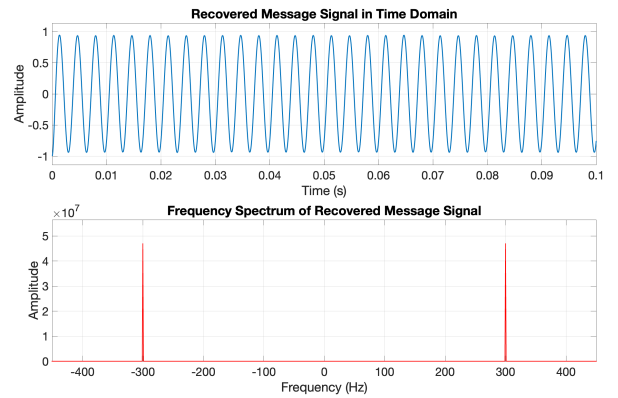


Figure 4: Spectrum of demodulated signal.

Since the time-domain signal was severely distorted due to noise, envelope detection could not be used. Instead, synchronous detection was employed as the carrier frequency was

known. The noisy signal was multiplied by the same carrier wave used for modulation, $\sin(2\pi f_c t)$.

A Butterworth low-pass filter of degree 2 with a cutoff frequency of 500 Hz (to eliminate contributions at $2f_c$) was then applied. After obtaining $A + m(t)$, the mean of the signal was subtracted to remove the DC component. The resulting signal was the original sine wave with a slightly reduced amplitude, likely due to the non-ideal characteristics of the Butterworth filter. The frequency spectrum confirmed the presence of Dirac pulses at ± 300 Hz, indicating successful recovery of the original message signal.

2. Demodulating a Voice Signal

After downloading and extracting the audio file, a time vector was created using the sampling frequency and the length of the audio signal.

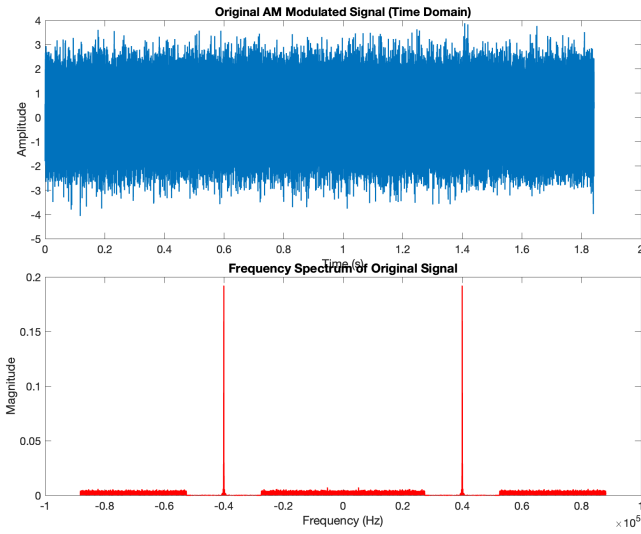


Figure 5: Original audio waveform.

While the time-domain waveform did not reveal clear information, the frequency domain analysis showed the carrier frequency of the modulated audio signal to be 40 kHz. Significant spectral noise density was observed across all frequencies except around 40 kHz. To address this, a bandpass filter with a central frequency of 40 kHz and a bandwidth of 5 kHz was applied. The bandpass filter clarified the signal and effectively removed interfering noise. The modulated signal was then multiplied by a carrier wave of 40 kHz, defined as $\cos(2\pi 40000t)$. A Butterworth low-pass filter with a cutoff frequency of 22 kHz (the audio signal bandwidth) and degree 10 was applied to eliminate higher frequency components.

Although the specified bandwidth was 22 kHz, a lower cut-off frequency (aligned with the human audible range, up to 10 kHz) was tested and found to further suppress noise. After removing the DC component and normalizing the signal for compatibility with audio formats, the signal was resampled to 44.1 kHz (the highest frequency allowed for MP3 format).

The final result resembled a clear audio signal in the time domain with zero DC component. In the frequency domain, the

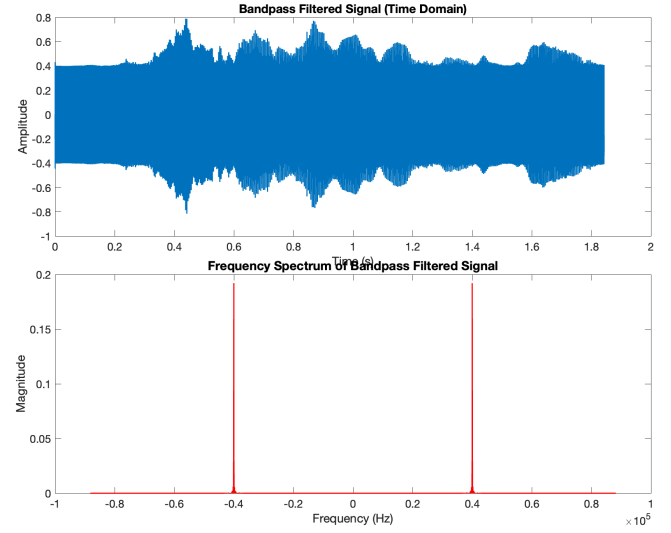


Figure 6: Bandpass filtered audio waveform.

practical bandwidth of the signal can be confirmed to be below 22 kHz from the FFT plot above. I concluded the project

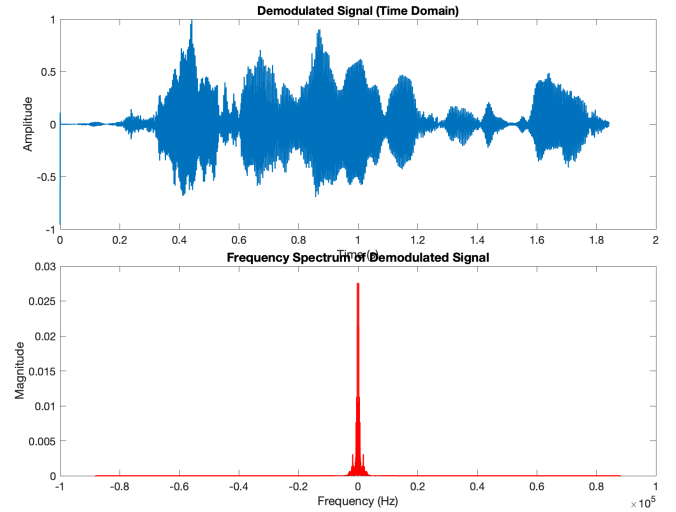


Figure 7: Demodulated audio waveform.

by successfully recovering and listening to a portion of one of Serdar Ortaç's famous songs, demonstrating the effectiveness of the demodulation process and the overall methodology.